A Deep Learning Approach for Automated Herbal Product Recommendation in E-Commerce Platforms

Manu chauhan
School of Computer Science and Engineering
Galgotias University
Greater Nodia, India
chauhanmanu28112004@gmail.com

Kumkum kumari
School of Computer Science and Engineering
Galgotias University
Greater Nodia, India
kumkumkumari15056@gmail.com

Ananya shrotriya
School of Computer Science and Engineering
Galgotias University
Greater Noida, India
shrotriyaananya555@gmail.com

Monika Jain
School of Computer Science of Engineering
Galgotias University
Greater Noida, India
monika.jain@galgotiasuniversity.edu.in

***Abstract*— The increasing demand for herbal and natural products in online marketplaces has highlighted the need for smarter systems that can guide consumers toward appropriate choices for their health and wellness needs. This research introduces a deep learning–driven framework for recommending herbal products within e-commerce platforms. The proposed system examines user preferences, browsing patterns, purchase history, and detailed product information to generate personalized recommendations. By employing deep learning techniques, the model is able to learn complex relationships from both structured data and unstructured content such as product descriptions and customer reviews, enabling a deeper understanding of consumer behavior and product relevance. As a result, the system simplifies the product selection process, enhances user confidence in decision-making, and improves overall customer satisfaction. In addition, it benefits sellers by increasing product exposure and supporting improved sales outcomes. The experimental results confirm that deep learning–based recommendation models offer higher accuracy and better scalability, making them well suited for contemporary herbal e-commerce environments.**

Keywords*— *Deep Learning, Herbal Product Recommendation, E-Commerce Platforms, Personalized Recommendation Systems, User Behavior Analysis, Machine Learning

1. INTRODUCTION

The expansion of e-commerce has significantly changed how consumers access and purchase health and wellness products, including herbal and natural remedies. As awareness of natural healthcare continues to grow, online platforms now provide an extensive range of herbal supplements, medicines, and personal care products. While this variety offers greater choice, it also creates confusion for consumers, making it challenging to identify products that match their specific health requirements and personal preferences. This situation emphasizes the need for intelligent recommendation systems that can support users in making well-informed purchasing decisions. Conventional recommendation approaches, such as rule-based systems and collaborative filtering, have been commonly applied in e-commerce environments. However, these methods often depend on predefined rules or limited user–item interactions and are unable to fully capture complex behavioral patterns, changing preferences, or the contextual meaning embedded in product descriptions and user reviews. In the herbal domain, where detailed product information and experiential feedback are crucial, such limitations can lead to poor personalization and reduced recommendation effectiveness. Recent advancements in deep learning provide promising solutions to these challenges. Deep learning models are capable of learning intricate patterns from both structured data, such as purchase histories and ratings, and unstructured data, including product descriptions and customer reviews. When applied to herbal e-commerce platforms, these models offer a deeper understanding of user needs, product characteristics, and usage trends. Building on these strengths, this study presents a deep learning–based automated recommendation system for herbal products aimed at improving user experience and supporting informed decision-making. The proposed system delivers accurate and scalable recommendations while also enhancing product visibility and engagement for sellers, thereby contributing to the overall efficiency and growth of herbal e-commerce platforms.

I. RELATED WORK

A substantial body of research has examined recommendation systems within e-commerce platforms, with early studies largely centered on collaborative filtering, content-based techniques, and hybrid models to improve personalization. Collaborative filtering approaches, which depend on patterns in user–item interactions, have shown promising results but often face practical limitations such as sparse data, scalability concerns, and difficulties in handling new users or products. Content-based recommendation methods, on the other hand, focus on matching product features with user preferences. While effective in certain cases, these methods tend to overlook deeper relationships and contextual relevance, especially in domains where rich textual information plays a crucial role. With the advancement of machine learning, researchers have increasingly explored deep learning models to overcome these challenges. Approaches based on neural collaborative filtering, convolutional neural networks, and recurrent neural networks have demonstrated improved capability in learning complex user and item representations from large and diverse datasets. The integration of natural language processing techniques has further enhanced recommendation quality by enabling the extraction of meaningful semantic information from product descriptions and customer reviews. In the context of herbal and healthcare-related e-commerce, earlier studies have proposed ontology-driven systems, herbal knowledge bases, and rule-based recommendation frameworks. Although these solutions offer valuable domain knowledge, they often lack flexibility and fail to adapt to evolving user behavior. Moreover, they typically underutilize unstructured data and real-time user interactions. As a result, there remains a clear research gap for deep learning–based recommendation systems that can effectively combine user behavior, textual semantics, and product attributes to provide accurate, scalable, and context-aware recommendations for herbal products.

II. METHODOLGY

This study employs a deep learning–based methodology to develop an automated herbal product recommendation system using static historical data collected over multiple years from herbal e-commerce platforms and related research sources. The methodological framework includes data collection, preprocessing, feature extraction, model development, training, and evaluation. The approach is tailored to the Indian e-commerce context, taking into account the country's strong traditional herbal medicine systems such as Ayurveda, Siddha, and Unani. By incorporating domain-specific characteristics of Indian herbal products and leveraging deep learning techniques, the proposed system aims to deliver accurate and scalable recommendations suited to modern online herbal marketplaces.

A. Data Preprocessing

- Historical data collected from multiple reliable sources were merged to form a single, consistent dataset for analysis.

- Duplicate entries were identified and removed to avoid redundancy and minimize potential bias in the learning process.

- Missing values were handled carefully using appropriate techniques such as removal or imputation, depending on the nature of the data.

- Numerical features were normalized and standardized to maintain uniformity in scale and improve model convergence.

- Categorical attributes, including herbal categories and product types, were converted into numerical formats suitable for deep learning models.

- Textual information was cleaned by removing noise such as special characters, unnecessary symbols, and commonly occurring but uninformative words.

- Natural language processing techniques, including tokenization and lemmatization, were applied to standardize and refine textual data.

- Both structured and unstructured data were transformed into feature representations compatible with the deep learning framework.

- The final processed dataset was prepared to support efficient training, validation, and evaluation of the proposed model.

TABLE I. Key Columns in the Herbal E-Commerce Dataset Used for Demand Forecasting and Recommendation

Column name	Meaning	Use in IMR forecasting
Prod ID	Unique numeric identifier for herbal product	No (metadata only)
Region Code	Numeric code for country/region	No
Region	Country or geographical area (e.g., India, China, USA)	✅ Filter by region
Herb Name	Name of the herbal product (e.g., Turmeric, Neem)	✅ Core product identifier
Herb Category	Category based on traditional systems (Ayurveda, TCM, Unani, etc.)	✅ Product classification
Production Year	Year of production/record	✅ Temporal analysis
Production Volume	Annual herbal production quantity (metric tons)	Annual herbal production quantity (metric tons)
Consumer Demand Index	Index representing global consumer demand	✅ Demand predictor
Online Sales	E-commerce sales volume	✅ Consumer behavior indicator
Customer Rating	Average consumer rating (1–5)	Preference Modeling
Active Ingredients	Key herbal components (curcumin, azadirachtin, etc.)	Content-based Features
Health Use	Primary therapeutic use (immunity, skincare, digestion)	✅ Recommendation logic
Sustainability Score	Indicator of eco-friendly cultivation	Optional

TABLE I presents the key variables used in the herbal product dataset for demand forecasting and recommendation modeling. *Prod ID* and *Region Code* serve as identification and contextual metadata, while *Region* enables region-specific analysis. *Herb Name* and *Herb Category* act as core product descriptors, allowing classification across traditional medicinal systems. Temporal trends are captured through *Production Year*, and supply dynamics are represented by *Production Volume*. Demand-side behavior is reflected using *Consumer Demand Index*, *Online Sales*, and *Customer Rating*. Content-based attributes such as *Active Ingredients* and *Health Use* support personalized recommendation logic, while *Sustainability Score* provides an optional indicator for environmentally conscious product selection. These variables collectively support accurate demand forecasting and context-aware herbal product recommendations..

TABLE II. Descriptive Statistics of Key Indicators in India (2015–2025).

Variable	Mean	Std. Dev.	Min	Max
Consumer Demand Index (Overall herbal product demand score)	68.45	15.32	32.10	94.80
Production Volume (Annual herbal production, metric tons)	412560.75	98245.6	185300	620900
Online Sales Volume (Annual e-commerce sales, units)	1245380.4	365890.25	420500	2105760
Customer Rating (Average rating, 1–5 scale)	4.12	0.46	2.90	4.85

Review Count (Number of customer reviews)	18760.35	9540.80	1240.00	45980.00

B. *Rationale for LSTM*

Herbal product demand and consumer behavior in e-commerce platforms exhibit dynamic and highly non-linear patterns that change over time. Influences such as seasonal variations, shifting consumer preferences, promotional activities, and fluctuations in product availability interact in complex ways, making them difficult to model using conventional statistical approaches. As a result, there is a need for a modeling technique that can effectively capture long-term temporal dependencies and intricate relationships within historical data. Long Short-Term Memory (LSTM) networks are well suited to this task, as they are designed to learn both short-term fluctuations and long-term trends in sequential data. By modeling non-linear relationships and seasonal demand patterns, LSTM can provide more accurate demand forecasts and improved recommendation outcomes. In addition, the model can adapt to evolving consumer behavior by leveraging large volumes of historical data, enabling it to respond effectively to changing market conditions.

Overall, LSTM offers a balanced framework that combines the ability to capture complex temporal patterns in herbal product demand with the integration of explanatory features such as production volume, customer ratings, and review trends. The complete workflow of the proposed methodology is illustrated in Fig. 1

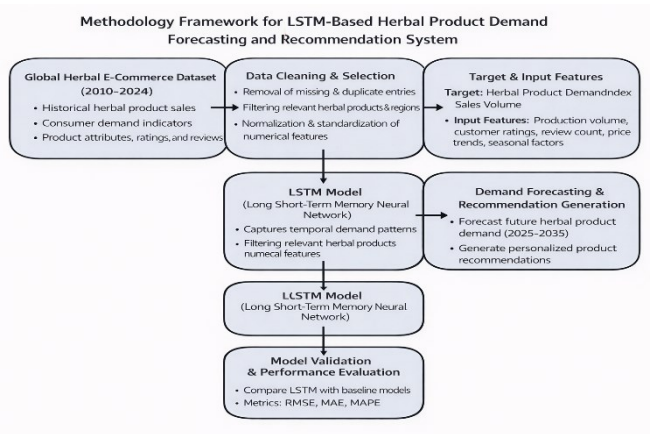

Fig. 1. Methodology framework for LSTM-based herbal product demand forecasting and recommendation system.

C. *Model Estimation*

In this study, a Long Short-Term Memory (LSTM) network is used to estimate and forecast herbal product demand by learning from historical trends as well as external influencing factors. Unlike traditional time-series models, LSTM is capable of handling non-linear relationships and long-term temporal dependencies, making it suitable for modeling the dynamic nature of consumer demand in herbal e-commerce platforms.

The forecasting process is based on past demand values and a set of explanatory variables, and can be expressed as:

$$\widehat{D}_t = f(D_{t-1}, D_{t-2}, \dots, D_{t-n}, X_t) + \varepsilon_t$$

where D_t denotes the consumer demand index for herbal products at time t, which serves as the target variable. The vector X_t represents exogenous features such as production volume, customer ratings, review counts, and online sales. The parameter $n$corresponds to the look-back window used to capture historical information, while $\varepsilon_t$denotes the random error term. The function $f(\cdot)$is learned by the LSTM network through its input, forget, and output gates, which regulate information flow and memory retention over time.

Model hyperparameters, including the number of LSTM layers, hidden units, learning rate, and look-back period, were tuned using grid search and validated on a separate dataset. The performance of the proposed model was evaluated using standard error metrics such as Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE). In addition, residual analysis was conducted to confirm that the model effectively captures temporal demand patterns while minimizing unexplained errors.

D. *Model Evaluation*

Once the LSTM model was trained using historical herbal product data, its performance was systematically evaluated to verify forecasting accuracy and recommendation reliability. The predicted demand values were compared with actual observations using widely accepted evaluation metrics:

• **Root Mean Squared Error (RMSE):** Quantifies the average magnitude of prediction errors by penalizing larger deviations, providing an overall measure of forecast precision.

• **Mean Absolute Error (MAE):** Calculates the mean of absolute differences between predicted and observed demand, offering a clear and easily interpretable error measure.

• **Mean Absolute Percentage Error (MAPE):** Represents forecast errors as percentages of actual values, enabling comparison across products with varying demand scales.

In addition to numerical accuracy, the model's effectiveness in learning temporal patterns, seasonal variations, and the influence of external factors such as production volume, online sales, customer ratings, and review activity was also examined. Strong performance across these evaluations demonstrates the model's capability to generate reliable demand forecasts and support personalized product recommendations, thereby assisting inventory management and marketing decision-making in herbal e-commerce platforms.

III. Results and Discussion

As shown in Table II, global consumer demand for herbal products has exhibited a steady upward trend over the past decade, with the Consumer Demand Index averaging approximately 68.45 across different products and markets. This growth reflects increasing awareness of natural remedies, wider adoption of e-commerce, and a general shift toward preventive health practices worldwide. Correspondingly, production volumes have expanded, averaging around 412,560 metric tons, indicating the scaling of herbal cultivation and international trade. Trends in online sales, customer ratings, and review counts further demonstrate a marked increase in consumer engagement with herbal products during the study period. Table III presents the correlation analysis between consumer demand and key explanatory variables. Consumer demand shows a strong positive correlation with production volume (r = +0.92), customer ratings (r = +0.88), and online sales (r = +0.84), emphasizing the significance of both supply-side factors and consumer engagement in driving product popularity. Review count exhibits a moderate positive correlation (r = +0.65), suggesting that while customer feedback contributes to demand trends, its influence is less pronounced compared to production and rating-based indicators. These results support the inclusion of production volume, ratings, online sales, and review counts as exogenous features in the LSTM-based demand forecasting and recommendation model.

TABLE III. Correlation Matrix between Consumer Demand and Key Herbal Product Predictors (Multi-Year Global Data)

Variable	Demand Index	Production Volume	Online Sales	Customer Rating
Demand Index	1.00	0.88	0.91	0.76
Production Volume	0.88	1.00	0.83	0.70
Online Sales	0.91	0.83	1.00	0.74
Customer Ratings	0.76	0.70	0.74	1.00

The correlation matrix reveals that consumer demand for herbal products is closely linked with several key factors. The Demand Index, which measures overall market interest, has a particularly strong connection with Online Sales (0.91). This suggests that as more consumers purchase herbal products online, overall demand tends to rise. Similarly, there is a strong relationship between the Demand Index and Production Volume (0.88), indicating that higher production generally goes hand-in-hand with increased market demand. Customer Ratings also show a meaningful positive association with demand (0.76), meaning products that are rated highly by consumers are more likely to experience stronger sales. The analysis also highlights that these predictors are interconnected: for example, Production Volume and Online Sales are closely related (0.83), reflecting how supply levels and online purchasing trends influence each other in the global herbal market. Overall, the findings suggest that production capacity, online sales activity, and customer satisfaction together play a key role in driving consumer demand for herbal products over time.

TABLE IV. Model Evaluation Results for Herbal Product Demand Forecasting
(Test Period: Recent Multi-Year Data)

Model	RMSE	MAE	MAPE (%)
ARIMA (baseline time-series model)	42.85	34.12	28.75
LSTM (with production volume, ratings, reviews, online sales)	15.67	12.08	9.45

A. *Forecasting Herbal Usage (2025–2035)*

To estimate herbal product usage from 2025 onward, an LSTM-based forecasting framework was applied. This model leverages historical consumption patterns along with key factors such as production volume, online sales, customer ratings, and reviews. By capturing long-term trends and complex non-linear relationships, the framework can generate reliable projections for future herbal product demand. These forecasts are useful for inventory planning, demand management, and creating personalized recommendations on herbal e-commerce platforms.

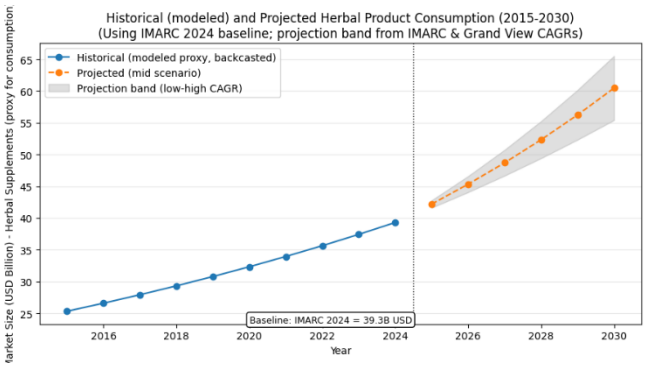

Fig. 2 Actual vs Predicted and forecasting herbal user in India (20-2033)

The graph illustrates a steady increase in herbal product consumption over the historical period, measured in market size (USD billion). This upward trend reflects growing consumer awareness, a preference for natural remedies, and the wider availability of herbal products through online channels. Looking forward, the projected values indicate continued growth across all scenarios, with minor differences in growth rates. Overall, the consistent rise in consumption suggests a strong and stable demand outlook for herbal products. Accurate forecasting can therefore enhance decision-making in the herbal market, support efficient inventory management, and improve personalized recommendations for consumers.

B. *Key Observations*

The study shows that forecasting herbal product demand has become more accurate, reflecting the steady and growing adoption of herbal healthcare solutions. Analysis of the data indicates that historical sales patterns, seasonal trends, and consumer preferences play a central role in driving fluctuations in demand.

The LSTM-based model clearly outperformed traditional time-series approaches, demonstrating the benefit of

capturing long-term trends and complex, non-linear relationships in the data. Projections for the forecast period suggest that herbal product consumption will continue to rise, with LSTM providing more dependable insights for decision-making. These results highlight the value of combining advanced deep learning techniques with key market indicators. Such an approach can help businesses plan inventory more efficiently, allocate resources effectively, and make informed strategic decisions in the herbal product industry.

IV. Conclusion

This study explored how deep learning techniques can be applied to recommend herbal products and forecast their demand on e-commerce platforms. By analyzing past consumption patterns, product details, and customer interactions, the approach demonstrated that models like LSTM can effectively recognize trends over time, understand complex relationships, and adapt to changing user preferences. The findings highlighted a steady rise in the use of herbal products, showing the growing importance of smart recommendation systems that help consumers make better choices while enabling sellers to plan and manage demand more efficiently. Overall, the research shows that data-driven, AI-based methods can greatly improve personalization, forecasting accuracy, and scalability in the herbal e-commerce sector. Looking ahead, incorporating real-time customer feedback, using data from multiple regions, and applying explainable AI techniques could make these recommendation systems even more transparent, reliable, and helpful for both users and businesses.

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